THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** rotation	about an axis
2	**(b)** periodic (circular/orbital)	repeats yearly
3	**(d)** linear motion	straight line
4	**(b)** hardness	e.g. diamond is hard
5	**(a)** solubility	e.g. salt dissolves
6	**(a)** Cotton shirt	plant fibre
7	**(c)** energy	definition
8	**(b)** expands when heated	thermal expansion
9	**(a)** mercury or coloured alcohol	expands uniformly
10	**(a)** larger	factor
11	**(c)** evaporation	surface process
12	**(c)** 0 °C	melting point

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Metre; kilometre.	1+1
14	Ability to dissolve in a liquid; salt dissolves; sand does not.	1+1
15	They are less dense than water.	1+1
16	It would show the temperature of the glass or burner.	1+1
17	Change from liquid to solid; water to ice in a freezer.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	At rest relative to the bus; in motion relative to the road (3).	1+2
19	Resistance to scratching (1.5); scratch one with the other (1.5).	1½+1½
20	Wash and shake it (1); keep it under the tongue or arm for a minute (1); read and clean it (1).	1+1+1
21	Temperature (1); surface area and wind (1); humidity (1).	1+1+1
22	Evaporation from oceans (1); condensation into clouds (1); rain returning water (1).	1+1+1
	OR: Water freezes; ice melts (3).	1+2
23	Rotation: turning about an axis, e.g. fan (1.5); oscillation: to-and-fro, e.g. swing (1.5).	1½+1½
	OR: Periodic: clock hands, pendulum, Earth (1.5); random: bird flying, leaf falling, children playing (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Utensils, wires (2). (ii) Sheets and wires (2). (iii) Foil and copper wire (1).	2+2+1
	OR: (i) On the basis of properties (1). (ii) Correct groups (3). (iii) Basis (1).	1+3+1
25	(i) Difference (2). (ii) Difference (2). (iii) Examples (1).	2+2+1
	OR: (i) States (2). (ii) Melting, freezing, evaporation, condensation (2). (iii) Diagram (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** at rest with respect to the bus stop	relative to reference
2	**(b)** rotational (circular) motion	turns about a fixed axis
3	**(b)** distance moved in unit time	definition
4	**(b)** Gas	spreads out
5	**(d)** Plastic	synthetic
6	**(a)** malleability	e.g. aluminium foil
7	**(b)** laboratory/room thermometer	range
8	**(b)** Boiling water	100 °C
9	**(d)** room temperature	approximate
10	**(b)** spread out in sunlight and wind	factors for evaporation
11	**(d)** boiling	bulk change
12	**(b)** Wet floor getting dry	liquid to gas

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Error due to viewing the scale at an angle; look straight above the mark.	1+1
14	Transparent lets light pass through; opaque does not.	1+1
15	It keeps the mercury level from falling after removal.	1+1
16	About 37 °C.	1+1
17	Change of vapour to liquid on cooling; dew drops.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Choose a reference point and give distance and direction (3).	1+2
19	Natural: cotton, wool (1.5); man-made: plastic, nylon (1.5).	1½+1½
20	Temperature measures hotness (1.5); heat is the energy transferred (1.5).	1½+1½
21	Pour water in a plate (1); leave in sunlight (1); water level falls and disappears (1).	1+1+1
22	Place the zero mark at one edge (1); keep the eye directly above the other edge (1); read the mark (1).	1+1+1
	OR: 0.36 m (3).	1+2
23	Air occupies space (inflated balloon) (1.5); has mass (1.5).	1½+1½
	OR: Metal is a good conductor (1.5); wood is an insulator (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Linear, circular, rotational, oscillatory, random (3). (ii) Periodic (1). (iii) Random (1).	3+1+1
	OR: (i) Procedure (2). (ii) 37 km (2). (iii) Coastline length (1).	2+2+1
25	(i) No (1). (ii) Clinical thermometer (1). (iii) Shake, place in the mouth, read (1). (iv) Consult a doctor (2).	1+1+1+2
	OR: (i) Precautions (2). (ii) Eye level (2). (iii) Touching the bottom, parallax (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** linear	straight line
2	**(b)** thread and a ruler	method
3	**(a)** handspans are not standard	non-standard
4	**(d)** mass	measured in kg
5	**(d)** Liquid	takes the shape of the container
6	**(a)** texture	rough or smooth
7	**(a)** a poor conductor of heat	insulator
8	**(b)** higher than normal	illness
9	**(c)** boiling water	its range is only 35–42 °C
10	**(c)** Evaporation of sweat takes heat from the body	cooling effect
11	**(a)** condensation of water vapour	breath has vapour
12	**(b)** frozen water in the form of ice crystals	precipitation

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Lay a thread along the curve, then straighten it and measure with a scale.	1+1
14	Resistance to scratching; diamond is hard, chalk is soft.	1+1
15	Clinical thermometer and laboratory thermometer.	1+1
16	Temperature and surface area (also wind, humidity).	1+1
17	Wind carries away the vapour so evaporation is faster.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Standard and accurate (3).	1+2
19	Rubber for tyres (flexible); glass for windows (transparent); steel for tools (hard) (3).	1+1+1
20	−10 °C to 110 °C; to measure many liquids (3).	1+2
21	Liquid to gas vs gas to liquid (1); heating vs cooling (1); takes vs gives heat (1).	1+1+1
22	Stone is denser than water (1.5); wood is less dense (1.5).	1½+1½
	OR: Lustre (1); malleability and ductility (1); good conductors (1).	1+1+1
23	Clinical: 35–42 °C and kink; laboratory: −10 to 110 °C no kink (1.5 each).	1½+1½
	OR: Not normal, a fever (1.5); consult a doctor and rest (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Density (1). (ii) Procedure (2). (iii) Wood, cork, wax float; stone, iron sink (2).	1+2+2
	OR: (i) Characteristics (1). (ii) Descriptions (2). (iii) Examples (2).	1+2+2
25	(i) Condensation (2). (ii) Droplets grow and fall (2). (iii) Snow, hail (1).	2+2+1
	OR: (i) Definition (1). (ii) Factors with activities (3). (iii) Examples (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.